

# Elliott Hirko

+1-352-251-3038 | [elliott.hirko@ufl.edu](mailto:elliott.hirko@ufl.edu)

 [www.linkedin.com/in/elliott-hirko-0941b22aa](https://www.linkedin.com/in/elliott-hirko-0941b22aa)

Gainesville, Florida

## EDUCATION

- **University of Florida** 2024 - 2027  
*Nuclear Engineering BS | GPA 3.32* Gainesville Florida
- **Santa Fe College** 2021 - 2024  
*Biomedical Equipment Technician AS* Gainesville Florida

## EXPERIENCE

- **University of Florida** Jan 2025 - Current  
*Undergraduate Research Assistant* Gainesville Florida
  - Working under Dr. Christopher McDevitt at the University of Florida utilizing physics informed neural networks to model electron runaway in Tokamak fusion reactors
  - Wrote physics informed neural network solver for the adjoint drift kinetic equation in order to model the effects of magnetic trapping on runaway probabilities
- **Santa Fe College** Feb. 2022 - Dec. 2023  
*Chemistry Lab Assistant* Gainesville Florida
  - Responsible for safety management in a dynamic laboratory environment, overseeing 50+ students.
  - Enforced rigorous lab etiquette and safety protocols.
  - Set up and operated standard analytical equipment, ensuring seamless lab operations.
  - Cleaned and organized equipment
  - Maintained equipment locker inventory

## PROJECTS

- **Physics Informed Nuclear Thermal-Hydraulic Analysis Code (PINTHAC)** September 2025 - Current  
*Python, Thermal Hydraulics, Data Analysis*
  - Building a Python library unifying various heat transfer models, machine learning libraries, and GPU-Acceleration methods
  - Enabling efficient low-level implementations of simple system analysis codes as well as routines for generating neural surrogates.
- **GPU-Accelerated IAPWS-95 Steam/Water Property Library** February 2026  
*PyTorch, Autograd, GPU Computing, Thermal Hydraulics*
  - Rebuilt the IAPWS-95 water/steam property formulation as a fully differentiable PyTorch library, exposing property derivatives via autograd instead of finite-difference approximations
  - Achieved 36x+ speedup over existing CPU-based property libraries by vectorizing evaluation across GPU tensors, enabling property lookups over entire spatial/temporal simulation grids in a single call
  - Designed as a drop-in differentiable backend for gradient-based thermal-hydraulic optimization and neural-network surrogate training
- **Liquid Metal Thermophysical Property Library** June 2026  
*Python, PyTorch, Uncertainty Quantification, Thermal Hydraulics*
  - Built a PyTorch-compatible library computing fluid and transport properties of liquid sodium, lead, and lead-bismuth eutectic (LBE) – the primary coolant candidates for Generation IV fast reactors
  - Implemented multiple literature-published correlations per property with tracked model-form uncertainty, enabling Monte Carlo error propagation in downstream thermal-hydraulic codes
  - Built to interoperate directly with the IAPWS-95 GPU library and the group's single-channel analysis codes, forming a shared property backbone across projects
- **Single-Channel Analysis Suite: FVM, FD & Neural Surrogate Solvers** October 2025 - Current  
*Python, PyTorch, Reactor Thermal Hydraulics, Neural Operators*

- Developed finite-volume single-channel analysis (SCA) codes with two-phase correlations to model PWR and BWR fuel-pin heat transfer, and extended the formulation to annular fuel geometry
- Wrote finite-difference and physics-informed neural network (PINN) solvers for transient single-channel analysis
- Trained a DeepONet surrogate that returns the full axial thermal profile of a fuel pin in real time from 10+ operating inputs and an arbitrary axial heating profile, replacing iterative solves with a single forward pass

#### • Inertial Electrostatic Confinement (IEC) Fusion Reactor

July 2022 - Current

*High Vacuum, High Voltage, Radiation Detection, Python*

- Designing and constructing a Farnsworth-Hirsch IEC fusion reactor, including a high-vacuum chamber system built from ConFlat/KF flanges and Viton bell-jar interfaces
- Designed and built a dedicated diffusion-pump control stand to operate the high-vacuum system
- Instrumented the device with Ludlum survey meters and a He-3 detector for radiation/neutron measurement, and developed Python models to predict device performance

#### • Undergraduate Chemistry Research

December 2021 - 2024

*Lab Skills, Organic and Inorganic Synthesis*

- Researched 3D printing of cellulose via stereolithography for tissue recellularization; secured \$1,000 in funding and mentorship through MIT's THINK program
- Synthesized and characterized organic/inorganic compounds (graphitic carbon nitrides, coordination complexes, fluorescent dyes) and tested their photocatalytic degradation performance

### SKILLS

- **Programming:** Python, Engineering Equation Solver (EES), OpenMC, LaTeX, moderate experience in Fortran and C
- **Machine Learning:** Experience writing neural networks in PyTorch
- **Scientific Writing:** Experienced in writing scientific reports including technical procedures, data compilation, and interpretation of results
- **Standard Lab Equipment:** Pipets, hotplates, balances, heating mantles, etc
- **Analytical Equipment:** Nicolet IS5 FTIR, Standard spectrophotometers, Nanalysis Benchtop NMR, Gamma Spectrometers (NaI(Tl) and HPGe)
- **Equipment Preventative Maintenance**
- **Fabrication:** Elementary operation of lathe and milling machine. Basic TIG welding

### HONORS AND AWARDS

#### • Deans List

2024-Current

*UF Wertheim Department of Engineering*

#### • United States National Chemistry Olympiad Qualifier

2023

*American Chemistry Society*

#### • MIT THINK Program Finalist

2023

*Massachusetts Institute of Technology*

#### • Outstanding innovation in STEM award from Santa Fe college

2023

*Santa Fe College*

#### • 3rd place in the Institute for Learning in Retirement's (ILR) Santa Fe College research exhibition

2023

*Institute for Learning in Retirement at Oak Hammock*

#### • 1st place in the Institute for Learning in Retirement's (ILR) Santa Fe college research exhibition

2022

*Institute for Learning in Retirement at Oak Hammock*